MANGALORE INSTITUTE OF TECHNOLOGY & ENGINEERING
(A Unit of Rajalaxmi Education Trust ®, Mangalore)
Autonomous Institute Affiliated to V.T.U, Belagavi, Approved by AICTE, New Delhi
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PROJECT PHASE – 1 REPORT

"Edge-Optimized ML Framework for Realtime Camouflaged Target Detection in Tactical Military Surveillance Systems"

Submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

in

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

by

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2025-26

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CERTIFICATE

This is to certify that the project work entitled **"EDGE-OPTIMIZED ML FRAMEWORK FOR REALTIME CAMOUFLAGED TARGET DETECTION IN TACTICAL MILITARY SURVEILLANCE SYSTEMS"** is a Bonafide work carried out by **CHAITHANYA R RAO (4MT23AI014), GOURAV (4MT23AI024), VINUSH KUMAR (4MT23AI061), YAKSHITH K D (4MT23AI062)** in partial fulfillment for the award of degree of **Bachelor of Engineering in Artificial Intelligence and Machine Learning** of **Mangalore Institute of Technology & Engineering (Autonomous)** during the year **2025-26**. It is certified that all corrections and suggestions indicated for Internal Assessment have been incorporated in the report deposited in the departmental library. The project has been approved as it satisfies the academic requirements in respect of project work prescribed for the Bachelor of Engineering degree.

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ACKNOWLEDGEMENTS

The successful completion of Phase – I of the Project titled “**Edge-Optimized ML Framework for Realtime Camouflaged Target Detection in Tactical Military Surveillance Systems**” would not have been possible without the support of those, whose constant guidance and encouragement helped turn our efforts into success.

We would like to thank our Project Guide **Dr. Maryjo M George**, Assistant Professor, Department of Artificial Intelligence and Machine Learning for her constant encouragement and support extended throughout.

We would like to express appreciation to **Dr. Sunil Kumar S,** Head of Department of **Artificial Intelligence and Machine Learning** for his support and guidance.

We would like to thank our Principal **Dr. Prashanth C M**, for encouraging us and giving us an opportunity to accomplish the project.

We also thank our management who helped us directly or indirectly in the completion of Phase – I of the project.

Our special thanks to faculty members and others for their constant help and support.

Above all, we extend our sincere gratitude to our parents and friends for their constant encouragement with moral support.

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ABSTRACT

Military surveillance and reconnaissance operations increasingly depend on AI-based vision systems to detect and track threats in complex natural environments. A critical and unsolved challenge in this domain is the detection of camouflaged objects — targets deliberately engineered to blend with their surroundings, making them nearly invisible to both human observers and standard detection algorithms. Existing high-accuracy Camouflaged Object Detection (COD) models are computationally intensive and reliant on cloud or GPU infrastructure, making them impractical for real-time deployment on compact, resource-constrained UAV platforms operating in communication-denied environments. This project proposes a lightweight, hardware-optimized COD framework designed for real-time inference on edge devices integrated into military drones. The objective of this system is to perform real-time camouflaged threat detection by utilizing a compressed deep learning model optimized through INT8 Quantization and Channel Pruning, and converted into a high-efficiency inference engine using TensorRT for deployment on the NVIDIA Jetson series of SoCs. The framework targets consistent real-time processing speeds exceeding 30 FPS while maintaining high structural precision through optimized S-measure scores and low Mean Absolute Error (MAE) in object boundary detection. The system incorporates robustness against adversarial camouflage techniques, including physical adversarial patches designed to deceive neural network-based detectors. Integration with existing UAV flight controllers is achieved through the MAVLink protocol, enabling plug-and-play deployment with minimal disruption to current hardware configurations. A key contribution of this system is its fully autonomous, cloud-independent operation — by moving all inference intelligence directly onto the drone, the framework eliminates satellite link latency, removes network security vulnerabilities, and enables reliable detection in remote and electronically contested zones. This project demonstrates the potential of edge-deployed deep learning in military reconnaissance by delivering a scalable, cost-efficient, and operationally autonomous solution for real-time camouflaged threat detection. In addition to eliminating dependence on cloud infrastructure and reducing network security risks, the framework provides a practical pathway for integrating AI-driven detection into battlefield UAV systems, laying the groundwork for future advancements in autonomous military vision systems.

Keywords: Camouflaged Object Detection, Deep Learning, Edge Computing, UAV, TensorRT, INT8 Quantization, Channel Pruning, Military Surveillance, MAVLink, Real-Time Inference

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Chapter 1

1. INTRODUCTION

1.1 Overview of the Domian:

Modern military surveillance increasingly relies on unmanned aerial vehicles (UAVs) and autonomous reconnaissance systems for real-time intelligence gathering, border monitoring, and tactical threat assessment [1], [2]. These systems operate in complex natural environments where hostile personnel, vehicles, and equipment are deliberately camouflaged to blend with surrounding terrain. Recent advances in deep learning, including CNNs, transformer-based architectures, attention-enhanced detection frameworks, and multispectral fusion models, have improved object detection under conventional conditions [1], 2].

However, Camouflaged Object Detection (COD) introduces unique and unsolved challenges that standard detection pipelines are not equipped to handle [3], [4]. Despite architectural progress, most high-accuracy COD models remain computationally intensive and dependent on cloud or GPU infrastructure, making them impractical for real-time edge deployment on compact drone platforms [5], [6]. This research focuses on developing a lightweight, hardware-optimized COD framework capable of real-time inference on resource-constrained edge devices integrated into military UAVs, without any cloud dependency [7], [8].

The key domain challenges this work addresses include:

- **High Structural Similarity:** Camouflaged objects exhibit minimal visual distinction from background textures, causing saliency noise, boundary ambiguity, and constrained receptive field failures in standard detection architectures [9], [10].
- **Adversarial Camouflage:** Modern stealth strategies have evolved beyond visual blending. Physical adversarial patches can simultaneously deceive optical and infrared detectors, while hyperspectral stealth coatings replicate environmental spectral signatures to evade AI-based detection entirely [11], [12], [13].
- **Edge Deployment Gap:** Existing COD research is primarily optimized for benchmark datasets and server-grade hardware, with limited emphasis on hardware-aware model compression, embedded deployment feasibility, and real-time video stream processing [2], [14], [15].
- **Cloud Dependency Risks:** Network-reliant processing introduces operational latency, increases security exposure, and fails entirely in communication-denied or remote battlefield environments [16].
- **Lack of System-Level Integration:** Current works rarely address end-to-end integration with automated response mechanisms, leaving a critical gap between laboratory-level COD performance and practical autonomous surveillance applications [17], [18].

1.2 Background:

Camouflage has been a fundamental element of warfare for centuries, but contemporary applications have made it far more sophisticated. Soldiers, vehicles, and equipment are now engineered to blend so completely into their surroundings that even trained human observers can fail to detect them [18]. This makes identifying camouflaged targets one of the most difficult problems in modern computer vision, with direct consequences for battlefield safety and mission success [3], [19].

The field of Camouflaged Object Detection has evolved through several generations of approaches. Early rule-based and handcrafted feature methods failed to generalize across terrain types, while first-generation deep learning models using standard CNN architectures were not specifically designed to handle the structural similarity between camouflaged targets and their backgrounds [20], [21]. More recent work with transformer-based architectures and attention mechanisms has improved boundary detection and global contextual reasoning, but at the cost of substantial computational overhead that makes deployment on resource-constrained hardware impractical [6] [22], [23].

Several factors define the current research landscape and motivate the direction of this project:

- **Evolving camouflage technology.** Modern camouflage now includes adversarial designs specifically engineered to confuse AI detection systems by exploiting known weaknesses in neural network architectures. As stealth technology advances, detection systems must keep pace [12], [18].
- **Fundamental limitations of existing AI models.** Even state-of-the-art deep learning models that achieve high benchmark performance frequently fail when objects are intentionally designed to resemble their background. Architectures built specifically to reason about structural gradients and texture-level differences are needed [10], [12].
- **The edge deployment gap.** Most high-accuracy COD models require powerful GPU hardware to operate, creating a significant gap between research progress and practical deployment on battlefield edge platforms such as micro-UAVs and wearable tactical systems [2], [7].
- **The cost of detection latency.** In time-critical military operations, delayed detection is operationally equivalent to no detection. A system requiring several seconds per frame cannot support real-time threat identification and response [1], [24].
- **Human risk reduction.** AI-based autonomous detection enables continuous monitoring of hostile environments where human presence is unsafe or infeasible, directly reducing the risk to reconnaissance personnel [25], [26].
- **Closing the research-to-deployment gap.** The majority of COD research concludes at benchmark accuracy, with very little work addressing efficient deployment on compact low-power hardware

[8], [14], [15]. This project directly targets that gap by combining detection performance with hardware-aware optimization.

1.3 Objectives:

The primary objective of this research is to develop a lightweight and hardware-optimized camouflage detection framework for real-time deployment in military drone and edge surveillance systems.

The specific objectives are:

1. To design a lightweight and computationally efficient deep learning model for accurate detection of camouflaged military personnel and vehicles in complex natural environments [2], [33].
2. To enable real-time inference on resource-constrained military edge platforms such as micro-UAVs, smart drones, and wearable tactical devices [21], [34].
3. To apply hardware-aware optimization techniques, including model compression, pruning, and quantization, to minimize computational overhead, latency, memory usage, and power consumption [17], [37].
4. To develop an onboard real-time video processing pipeline that performs frame extraction, camouflage detection, and immediate threat alert generation directly on the edge device [9], [34].
5. To implement an autonomous trigger-based response mechanism that reduces dependency on centralized command servers or cloud infrastructure [18], [39].

Chapter 2

2. LITERATURE SURVEY

The literature survey is organized across five methodological domains aligned with the proposed system's core technical objectives: Camouflaged Object Detection Architectures, Lightweight Model Compression and Quantization, Edge Deployment and Embedded Inference Optimization, Adversarial Camouflage and Robustness Techniques, and UAV-Based Real-Time Vision Systems. All 17 papers selected are implementation-based studies (not surveys or theoretical analyses) published between 2022 and 2025 in venues including IEEE Xplore, Springer, and EBSCO.

Table 2.1 Literature Survey

S.No	Paper Title	Methodology	Limitations	Research Gaps Identified	Year
1	Camouflaged Object Detection Using Hybrid-Deep Learning Model	Swin Transformer backbone combined with Enhanced Receptive Field (ERF) and CrossScale Feature Fusion.	High computational overhead; strictly RGB-dependent; slow inference speeds.	Need for lightweight real-time systems, multimodal fusion (RGB-thermal), and video validation.	2025
2	Hybrid-Recursive-Refinement Network for Camouflaged Object Detection	CNN-Transformer hybrid encoding with a recursive decoder refinement mechanism.	Recursive refinement drastically spikes inference costs; untested on video or edge devices.	Efficient real-time models and robust extension to edge-device scenarios.	2025
3	Identification of Camouflage Military Individuals with Deep Learning Approaches DFAN and SINetV2	Multi-view feature generation (DFAN) coupled with texture enhancement and group-reversal attention (SINet-V2).	Computationally heavy multi-view design; web-sourced datasets introduce bias.	Requirement for lightweight architectures and unbiased, robust military video datasets.	2025

4	Camo Vision : A Dual-Mode Deep Learning Framework for Camouflaged Object Detection in Images and Videos	U-Net encoder decoder with ResNet-50 backbone using frame-wise Exponential Moving Average (EMA).	Frame-wise processing lacks true temporal memory; prone to overfitting on military scenes.	True temporal learning models, multimodal fusion, and broader cross domain evaluation.	2025
5	Military Camouflaged Target Detection Through Multimodal Images: A Comparative Study	Independent YOLO-based evaluation across Visible (VIS), Near-Infrared (NIR), and Long Wave Infrared (LWIR) bands	Treats the detector as a black box; absolute lack of multimodal feature fusion	Absence of feature level or decision-level fusion and adaptive cross-modal attention.	2025
6	Camouflaged Target Detection Based on Multimodal Image Input Pixel-Level Fusion	Introduces MIF-YOLOv5 for direct pixel-level fusion of optical (RGB) and Infrared (IR) data.	Direct fusion causes "ghosting" artifacts during high-speed sensor motion; computationally heavy.	Robustness against electronic countermeasures; need for adaptive fusion weighting based on lighting.	2024
7	Camouflaged Military Personnel Detection Using SSIM-Based Faster R-CNN Model	Two-stage Faster R-CNN augmented with an SSIM-based perceptual loss component.	Two-stage detection is too slow for real-time video; potential overfitting to training textures.	Need for transformer-based context modeling and scene-level reasoning modules.	2025
8	ATDMNet: Multi-Head Agent Attention and Top-k Dynamic Mask for Camouflaged Object Detection	Uses Multi-Head Agent Attention (MHA) and Top-k Dynamic Masking (TDM) for cross-scale fusion.	Multi-scale masking remains too computationally heavy for low-power edge CPU deployment.	Solving fragmented segmentation and global-local trade-offs without excessive compute.	2025

9	Improved Camouflage Personnel Detection Based on YOLOv8	Modifies YOLOv8with a C2f-TA backbone,Triplet Attention (TA), and BiFPN multiscale fusion	High computational requirement; lacks mathematical explanation for TA; zero edge deployment analysis.	Video-based detection, real-time edge deployment profiling, and lightweight transformer integration.	2023
10	MCOD: The First Challenging Benchmark for Multispectral Camouflaged Object Detection	Introduces a multispectral dataset explicitly focused on detecting sub-1% extremely small objects.	Heavy model dependency; limited baseline comparison restricted only to YOLO variants.	Lightweight multispectral architectures optimized specifically for edge deployment.	2024
11	SAM2-DFBCNet: A Camouflaged Object Detection Network Based on the Hiera Architecture of Segment Anything Model 2	Utilizes the massive SAM2 (Hiera) backbone for extensive boundary and feature refinement.	Massive computational cost makes embedded military edge feasibility impossible.	Need for highly distilled, lightweight transformer segmentation models.	2025
12	EdgeYOLO: An Edge-Real-Time Object Detector	Employs an anchor-based grid detector with a C2f/C3 backbone optimized strictly for inference speed.	General-purpose design lacks camouflage-specific structural similarity suppression.	Camouflage-specific edge models integrating Triplet Attention and lightweight hybrid transformers.	2023

13	Ulcod-net: an ultralightweight camouflage object detection framework with gated multi-level feature fusion and dual constraint refinement	Uses a MobileViT backbone, Lightweight Boundary-Region Decoder, and gated attention.	"Lightweight" claims rely on high-end GTX 4090 GPUs; moderately heavy for actual ARM CPU inference.	True hardware-aware optimization (quantization/pruning) validated on ARM/Raspberry Pi.	2025
14	AINet: Integrating Mamba and CBAM for Enhanced Camouflage Object Detection	CNN/transformer backbone integrating Mamba, CBAM, and progressive decoder fusion.	Increased architectural complexity via multiple refinement modules; completely lacks temporal modeling.	Motion-aware camouflage detection, real-time deployment, and field domain generalization.	2026
15	Object Detection on Real-Time Video with FPN and Modified Mask RCNN Based on Inception-ResNetV2	Multi-scale video frame extraction using Inception-ResNetV2 and FPN within Mask R-CNN.	Processes frame-wise without temporal memory on an excessively heavy backbone.	Lightweight temporal feature fusion for smooth, motion-aware video tracking.	2024
16	CPANet: Cross-Domain Polynomial Attention Network for Camouflaged Object Detection	Dual-backbone architecture using FFT for frequency decoupling and KAN for nonlinear enhancement.	Fixed polynomial orders limit scene adaptability; struggles heavily with local fine-grained structures.	Adaptive polynomial orders and lightweight local frequency modeling.	2026
17	Camouflage soldier object detection network based on the attention mechanism and pyramidal feature shrinking	YOLOv7 backbone with CBAM, Pyramidal Feature Shrinking (FS-FPN), and an Information Handle Module.	Relies on an outdated, heavy backbone; restricted to static image, single-class detection.	Cross-dataset generalization, temporal video modeling, and lightweight drone architectures.	2024

2.1 Key Insights:

Analysis of the 17 reviewed papers reveals several consistent patterns and insights that directly inform the design of the proposed system:

- **Accuracy-efficiency trade-off dominates COD research.** Every paper achieving high benchmark accuracy does so at the cost of computational complexity that precludes edge deployment. No existing work achieves both simultaneously.
- **Edge deployment is consistently identified as a research gap.** Twelve of the seventeen reviewed papers explicitly acknowledge real-time edge deployment as an unresolved gap, yet none address it in their contributions.
- **Gradient and boundary supervision improves detection.** Papers using structural gradient supervision, such as ATDMNet and the SSIM-guided Faster R-CNN, achieve better boundary detection than purely semantic approaches.
- **Multimodal data improves robustness.** Studies using combined RGB, NIR, and LWIR data show improved detection in adverse conditions, but the fusion architectures are too heavy for embedded platforms.
- **Model compression is underexplored in COD.** Only ULCOD-Net and EdgeYOLO attempt any form of compression, and neither validates performance on actual ARM or embedded hardware under real operational constraints.

2.2 Research Gap Identification:

Based on the comprehensive review, the following critical research gaps have been identified:

- **No existing COD model is validated for real-time deployment on military-grade embedded hardware** (NVIDIA Jetson, ARM SoC) under actual power and memory constraints.
- **Adversarial camouflage robustness has not been integrated into any lightweight COD pipeline.** Existing robustness work operates on server-grade hardware only.
- **End-to-end autonomous UAV integration is absent from the literature.** No reviewed paper addresses the complete pipeline from edge inference to MAVLink-based flight controller communication.
- **Hardware-aware quantization and pruning are not applied to COD-specific architectures.** General compression tools have been applied to generic object detectors, but not to architectures designed for camouflage boundary detection.

Chapter 3

3. PROBLEM DEFINITION

3.1 Problem Statement:

"Existing camouflaged military target detection models achieve strong accuracy using heavy Transformer-based architectures, but their high computational cost and latency make them impractical for deployment on resource-constrained edge platforms, creating the need for an efficient, low-power, real-time edge-native detection framework." [1], [3], [15]

Existing machine learning models for detecting camouflaged military personnel and vehicles primarily rely on computationally intensive architectures such as Transformer-based [15] networks and heavy backbone designs, including models inspired by SSIM-guided and attention-driven frameworks. While these state-of-the-art approaches achieve high accuracy in benchmark environments, they require substantial GPU resources and involve complex processing pipelines, resulting in high inference latency.

Such models are not suitable for deployment on military edge platforms — including AI-enabled micro-UAVs, smart drones, wearable tactical systems, and smart helmets — which operate under strict constraints of limited memory, low power availability, and lightweight embedded processors. The high computational overhead and storage requirements of existing models prevent real-time inference and autonomous operation in battlefield conditions.

Therefore, there is a critical need for an edge-native camouflage detection framework that reduces computational complexity while maintaining detection reliability. This requires the integration of model compression, quantization, architectural pruning, and hardware-aware optimization [12], [16] techniques to achieve higher frame rates, lower latency, and minimal power consumption without compromising operational accuracy on embedded military devices.

3.2 Scope of the project:

This project focuses on the design, development, and edge deployment of a lightweight camouflaged object detection framework tailored for military UAV and tactical surveillance applications. The scope is defined as follows:

- **Detection Target:** Camouflaged military personnel, vehicles, and equipment in complex natural environments including forests, desert terrain, and grasslands.
- **Model Design:** Development of a lightweight deep learning architecture (DGNet-based) optimized for camouflage detection, incorporating hardware-aware compression techniques including INT8 quantization, channel pruning, and backbone optimization.

- **Edge Deployment:** The framework will be deployed on NVIDIA Jetson-series SoCs using TensorRT-optimized inference engines, targeting real-time performance exceeding 30 FPS without cloud or server dependency.
- **Onboard Pipeline:** An end-to-end video processing pipeline will perform frame extraction, inference, and threat alert generation directly on the edge device.
- **Autonomous Response:** A trigger-based autonomous alert mechanism will be implemented to reduce dependency on centralized command infrastructure.
- **UAV Integration:** Integration with UAV flight controllers via the MAVLink protocol, enabling seamless embedding into existing drone systems.
- **Out of Scope:** Hardware manufacturing, custom SoC design, satellite communication systems, and non-military civilian surveillance applications are outside the scope of this project.

3.3 Constraints and Assumptions:

3.3.1 Constraints:

1. Dataset Constraints

- Limited availability of high-quality camouflaged object datasets.
- Existing datasets may not cover all real-world terrains, weather conditions, and camouflage patterns.
- Small object instances and heavily blended targets are difficult to annotate accurately.
- Dataset imbalance may occur due to fewer camouflaged targets compared to background regions.

2. Computational Constraints

- Deep learning-based COD models require high computational resources for training and inference.
- Real-time detection can be difficult on low-power or edge devices.
- Large backbone networks increase memory usage and inference latency.

3. Environmental Constraints

- Detection performance may degrade in:
- Dense vegetation
- Complex textured backgrounds

- Low illumination environments
- Fog, rain, smoke, or shadows
- Dynamic backgrounds and background-object similarity make segmentation challenging.

4. Model Constraints

- Small, partially visible, or highly blended objects are difficult to detect.
- False positives may occur when background textures resemble object boundaries.
- COD models may struggle to generalize to unseen camouflage patterns.
- Fine edge extraction and boundary localization remain challenging.

5. Operational Constraints

- Real-time surveillance systems require low latency and high reliability.
- Edge deployment may be constrained by battery power and hardware limitations.
- Continuous monitoring systems require stable and optimized processing pipelines.

3.3.2 Assumptions:

1. Data Assumptions

- Input images are assumed to be properly preprocessed and normalized.
- Training data is assumed to contain representative camouflage scenarios.
- Images are assumed to have sufficient resolution for feature extraction.

2. System Assumptions

- The target object contains subtle distinguishable patterns, edges, or texture variations from the background.
- The camera system provides stable image acquisition.
- GPU acceleration is available during model training.

3. Environmental Assumptions

- The operational environment allows visible feature extraction from the scene.
- Extreme environmental distortions are assumed to be limited during standard operation.
- Objects remain partially visible within the scene.

4. AI/Model Assumptions

- Deep neural networks can learn hidden texture and contextual differences between object and background.
- Attention mechanisms improve focus on concealed regions.
- Multi-scale feature extraction improves small object detection capability.

5. Deployment Assumptions

- The system is intended for surveillance, reconnaissance, and monitoring applications.
- Trained models can be updated or fine-tuned when new camouflage patterns are encountered.
- The deployment platform supports the required inference framework and dependencies.

Chapter 4

4. PROPOSED METHODOLOGY

4.1 Proposed Approach and Algorithm:

The proposed system adopts a Deep Gradient Network (DGNet) [18] as its core detection architecture. DGNet is specifically designed for Camouflaged Object Detection and addresses the fundamental limitation of standard detectors — their reliance on color and texture features that camouflage explicitly mimics. DGNet decouples the detection process into two complementary streams: Contextual Awareness and Texture Gradient Identification, enabling the system to effectively detect camouflaged targets even when thermal signatures are suppressed or unavailable.

The detection framework operates as follows:

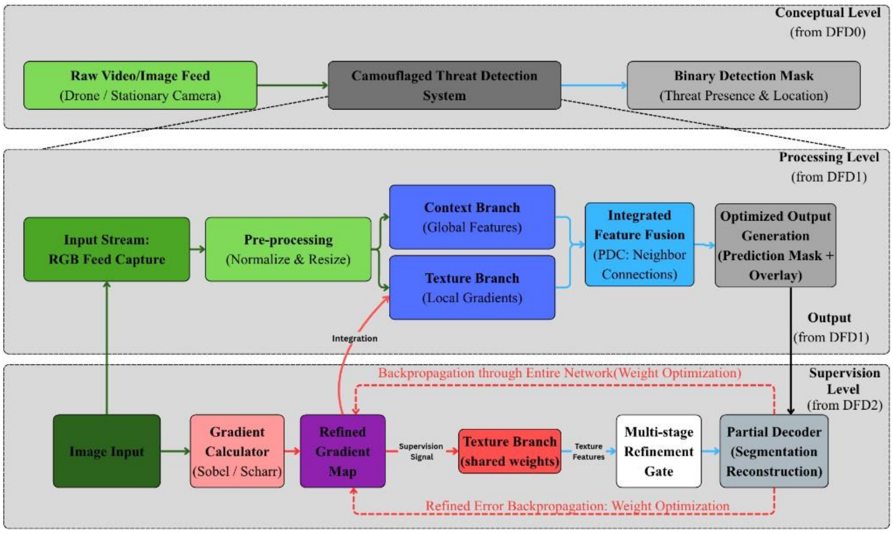

Fig 4.1 DGNet Camouflaged Threat Detection

- **Context Branch (The Searcher):** Uses a deep backbone (EfficientNet or ResNet) to capture global semantic features, answering the question: Where is the general area of interest?
- **Texture Branch (The Identifier):** A specialized branch that monitors Gradient-Induced Transitions (GIT) [18]. It acts as a structural probe to find the unnatural edges of military hardware, rifles, or personnel by supervising the network to identify sharp changes in pixel intensity.
- **Partial Decoder Component (PDC):** A unified decoder aggregating multi-level features from both branches using neighbor connections to preserve high-resolution spatial details, resulting in a precise binary detection mask [18].

The model implements a decoupled learning strategy where the context and texture are treated as complementary tasks.

Mathematical Formulation

The Image Gradient ∇I is the vector of partial derivatives [18]:

$$\nabla I = [G_x, G_y] = \left[\frac{\partial I}{\partial x}, \frac{\partial I}{\partial y}\right] \quad \text{—————— (1)}$$

I= Input image
$\frac{\partial I}{\partial x}$= Partial derivative of image intensity along x-axis
$\frac{\partial I}{\partial y}$= Partial derivative of image intensity along y-axis
∇I= Image gradient vector

The Gradient Magnitude M used for supervision is :

$$M = \sqrt{G_x^2 + G_y^2} \quad \text{—————— (2)}$$

G= Gradient magnitude
G_x= Horizontal gradient
G_y= Vertical gradient

The overall Objective Function L_{total} is a weighted sum of the Binary Cross-Entropy (BCE) loss and a specific Gradient Loss (L_{grad}):

$$L_{total} = L_{BCE} + \lambda L_{grad} \quad \text{—————— (3)}$$

L_{total}= Final objective loss
L_{BCE}= Binary Cross-Entropy loss
$L_{gradient}$= Gradient loss
λ= Balancing hyperparameter
G_{pred}= Predicted gradient map

Where:

$$\boldsymbol{L_{gradient} = | G_{pred} - G_{gt} |} \text{--------------------(3.1)}$$

G_{gt}= Ground truth gradient map
$L_{gradient}$= Gradient supervision loss
This improves the detection of highly blended camouflaged objects.

$$L_{BCE} = -\frac{1}{N}\sum_{i=1}^{N}[y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)] \text{----------------} (3.2)$$

N= Total number of pixels
y_i= Ground truth label of pixel i
$\hat{y}_i$= Predicted probability of pixel i
L_{BCE}= Binary Cross-Entropy loss

4.2 Techniques and Tools

4.2.1 Model Compression and Optimization

- **INT8 Quantization:** Reduces model precision from 32-bit floating-point to 8-bit integer representation, significantly reducing memory footprint and inference time with minimal loss in detection accuracy.
- **Channel Pruning:** Removes redundant filters from convolutional layers to reduce computational cost while preserving detection performance through structured sparsity.
- **TensorRT Conversion** [16]**:** PyTorch-trained models are exported to ONNX format and converted to TensorRT inference engines for GPU-accelerated deployment on NVIDIA Jetson hardware.
- **Knowledge Distillation:** A lightweight student network is trained to replicate the behavior of the larger teacher model, enabling further parameter reduction.

4.2.2 Software and Hardware Requirements:

Hardware Requirements:

Table 4.2.1: Hardware Requirements

Category	Minimum Requirement	Recommended Requirement
Processor	Intel Core i5 (2.4 GHz) or equivalent	Intel i7/i9 or AMD Ryzen 7/9
RAM	8 GB DDR4	16 GB or higher (for large datasets)
Storage	20 GB available space	512 GB SSD (for model storage & caching)
GPU	NVIDIA GTX 1650 (4GB VRAM)	NVIDIA RTX 3060+ (8GB+ VRAM)
Display	1080p resolution	4K resolution (for detailed analysis)
Network	100 Mbps Ethernet	1 Gbps (for real-time video processing)

Software Requirements:

Table 4.2: Software Requirements

Category	**Specifications**
Operating System	Windows 10/11, Ubuntu 18.04+, macOS 12+
Python Version	3.8 or higher
Web Framework	FastAPI0.104+ with Uvicorn server
Deep Learning Frameworks	PyTorch (Training), TensorFlow (Training + TFLite conversion)
ML Frameworks	OpenCV, NumPy
Edge Deployment Platforms (Future)	Ubuntu (Linux), Raspberry Pi OS, NVIDIA JetPack SDK
Additional Libraries	Pillow, Matplotlib, SciPy
Web Browser	Chrome 90+, Firefox 88+, Safari 14+
Development IDE	VS Code, PyCharm, Jupyter Notebook

4.2.3 Development Frameworks

- **PyTorch:** Primary framework for model design, training, and experimentation. Dynamic computational graph enables rapid prototyping and debugging.
- **TensorFlow Lite (TFLite):** Used for model conversion and deployment on Android and Raspberry Pi-class devices, supporting INT8 quantization and NNAPI hardware acceleration.
- **ONNX** [16]**:** Intermediate representation format enabling model portability between PyTorch and deployment runtimes including TensorRT and OpenVINO.
- **OpenVINO:** Intel toolkit for optimized inference on CPU/VPU-based edge hardware, used for Intel-based deployment targets.
- **MAVLink Protocol:** Lightweight messaging protocol used to communicate detection alerts and coordinates from the edge inference engine to UAV flight controllers.

4.3 System design and Workflow:

4.3.1 Data flow design:

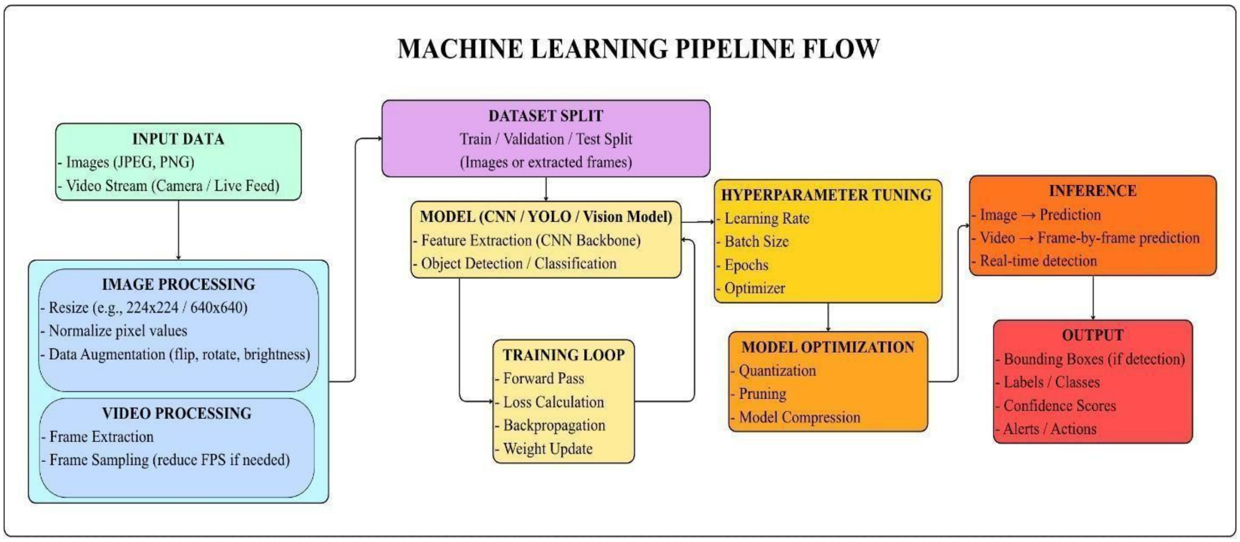

Fig 4.2 Data flow design

The data flow operates as a sequential machine learning pipeline, progressing from raw data ingestion to optimized real-time inference.

- Data Ingestion (The "Input Stage"): Processes static images and live video streams through dimensional resizing, normalization, spatial augmentation, and frame extraction.
- Model Training (The "Learning Core"): Partitions data into subsets and trains the feature extraction architecture, optimizing accuracy via hyperparameter tuning and iterative weight updates.
- Real-time Inference (The "Deployment Engine"): Compresses the network using quantization and pruning for edge devices, executing frame-by-frame detection to generate bounding boxes and automated alerts.

4.3.2 Deployment Design: Cloud vs. Edge:

For tactical military use, the project prioritizes **Edge Deployment**:

- **Edge Hardware:** NVIDIA Jetson Orin or specialized AI accelerators on edge devices or raspberry pi 5
- **Optimization:** The model is converted to **ONNX** or **TensorRT** formats to achieve **80+ FPS.**
- **Architecture:** A containerized approach using **Docker** ensures that the AI service can be updated over the air (OTA) in the field.
- **API Integration:** A lightweight FastAPI server is used to communicate detection coordinates to the Command and Control (C2) interface.

4.3.3 Advantages of proposed Design:

- **Counter-Stealth Capability:** Specifically targets the physical texture anomalies that modern camouflage cannot fully hide.
- **Thermal Independence:** Remains functional even if thermal sensors are jammed, blocked by thermal blankets, or unavailable due to dataset constraints.
- **Real-Time Efficiency:** The lightweight DGNet-S variant is optimized for high-speed reconnaissance.
- **Robust Boundary Detection:** Outperforms standard semantic models by providing sharp, clear edges of identified targets.

4.3.4 Hypothesis and Evaluation Matrix:

- **Null Hypothesis (H_0):** The integration of Gradient-Induced Transitions (GIT) does not provide a statistically significant improvement in the detection accuracy of camouflaged objects compared to traditional semantic segmentation models.

- **Alternate Hypothesis (H_1):** Explicit supervision of the texture gradient branch significantly reduces pixel-wise error and improves target boundary localization in camouflaged scenarios.

4.3.4 Performance Metrics:

- **Precision/Recall/F1-Score:** Standard measures for detection accuracy.
- **Mean Absolute Error (MAE):** Measures the average pixel-wise difference between the prediction and the ground truth.
- **Structure-measure (S_m):** A specialized metric that evaluates the structural similarity between the predicted mask and the target (crucial for military identification).

- **E-measure (E_m):** Evaluates both pixel-level matching and image-level statistics to capture target-background contrast.

5. SUMMARY AND FUTURE PLAN

5.1 Summary of Work Completed in Phase 1:

Phase 1 of this project has established the complete theoretical and architectural foundation for the Edge-Optimized ML Framework for Real-time Camouflaged Target Detection in Tactical Military Surveillance Systems. The following activities have been completed:

- **Literature Survey:** A comprehensive review of 17 implementation-based research papers (2022–2026) was conducted across five technical domains: COD architectures, lightweight model compression, edge deployment optimization, adversarial camouflage robustness, and UAV-based real-time vision systems. Critical research gaps were identified and documented.
- **Problem Definition:** A precise, technically grounded problem statement was formulated. The scope, constraints, and assumptions governing the system were defined, establishing clear boundaries for Phase 2 implementation.
- **Architecture Selection:** The Deep Gradient Network (DGNet) was selected as the core detection architecture based on its gradient-supervised dual-branch design, which directly addresses the structural similarity challenge of camouflaged targets.
- **Methodology Design:** The complete three-stage pipeline — data ingestion and preprocessing, model training and optimization, and edge inference with autonomous response — was designed and documented.
- **Technology Stack Definition:** The complete set of tools, frameworks, and hardware platforms for training, optimization, and deployment has been identified: PyTorch, TensorRT, ONNX, OpenVINO, NVIDIA Jetson Orin, and MAVLink.
- **Dataset Planning:** Three primary COD training datasets (COD10K, NC4K, CAMO) have been identified along with a plan for synthetic tactical data generation to supplement training for military-specific environments.

5.2 Plans for Phase 2 Implementation:

Phase 2 will transition from design to full implementation, targeting the following milestones:

1. Dataset preparation and augmentation: Download and preprocess COD10K [19], NC4K [20], and CAMO [19] datasets. Generate synthetic military environment data for jungle, desert, and urban terrain. Apply augmentation pipeline including Gaussian noise, multi-scale cropping, and flipping.
2. DGNet model training [18]: Train the DGNet model with gradient-supervised dual-branch architecture on the prepared datasets. Optimize the λ hyperparameter balancing BCE and gradient losses through systematic evaluation.

3. Model compression: Apply INT8 quantization and channel pruning to the trained model. Validate that detection accuracy is preserved within acceptable limits (less than 3% MAE degradation) post-compression.
4. TensorRT engine conversion and Jetson deployment: Export the compressed model to ONNX format, convert to a TensorRT inference engine, and deploy on the NVIDIA Jetson Orin target platform.
5. Real-time pipeline integration: Implement the live video feed processing pipeline with frame extraction, inference, bounding box overlay, and automated alert generation on the edge device.
6. Performance evaluation: Benchmark the complete system against defined metrics: S-measure [22], E-measure [21], MAE, Precision, Recall, F1-Score, FPS on Jetson hardware, power consumption, and model size. Validate against the null and alternate hypotheses defined in the system design.

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7. APPENDICES

APPENDIX A- DATASET OPERATIONAL MAPPING

TABLE I DATASET ROLE AND EVALUATION MAPPING

Dataset	Role	Focus
COD10K	Training	Texture-gradient feature learning
CAMO	Training	High visual-similarity extraction
NC4K	Testing	Boundary localization robustness
Synthetic	Tuning	Domain-specific terrain simulation

APPENDIX B- SYSTEM SPECIFICATIONS AND ENVIRONMENT

TABLE II HARDWARE AND SOFTWARE CONFIGURATIONS

Component	Specification
Processor	Intel i7/AMD Ryzen 9 (Training); Jetson Orin (Edge)
Memory	16 GB DDR4
GPU	NVIDIA RTX 3060+ (8 GB VRAM)
OS	Ubuntu 20.04 LTS / Windows 11
Frameworks	PyTorch, TensorFlow Lite, TensorRT
Libraries	OpenCV, NumPy, ONNX, MAVLink

APPENDIX C - OPTIMIZATION AND COMPRESSION METHODOLOGIES

- **INT8 Quantization**: Conversion of 32-bit floating-point weights to 8-bit integers to reduce memory footprint and latency.
- **Channel Pruning**: Removal of redundant convolutional filters to reduce FLOPs and improve inference speed.
- **Knowledge Distillation**: Training a compact student network to replicate the feature extraction of a larger teacher model.
- **TensorRT Conversion**: Compiling models into hardware-specific engines for NVIDIA Jetson platforms.

APPENDIX D - EVALUATION METRICS

- **Precision (P)**: Ratio of true positive pixels to total predicted positive pixels.
- **Recall (R)**: Ratio of true positive pixels to total ground truth positive pixels.
- **F1-Score**: Harmonic mean of **P** and **R**.
- **Mean Absolute Error (MAE)**: Pixel-wise average discrepancy between prediction and ground truth.
- **S-measure**: Structural similarity evaluation for object and region-aware assessment.
- **E-measure**: Alignment evaluation combining local pixel values and global image features.

APPENDIX E - SYSTEM OUTPUTS AND RESEARCH ROADMAP

System outputs include real-time bounding boxes, binary segmentation masks, and autonomous threat alerts via MAVLink. Future research focuses on multispectral sensor fusion, adversarial robustness, and decentralized swarm coordination for collaborative reconnaissance.

APPENDIX F - PSEUDOCODE / ALGORITHM STEPS

```
START

Capture live UAV video stream

FOR each frame:
    Preprocess frame
    Run DGNet inference
    Detect camouflage target
    Generate bounding box
    Estimate threat confidence

IF threat detected:
    Send MAVLink alert
    Trigger autonomous response

END
```